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Scientific background document

Leveraging AI tools for a strengthened biosecurity framework

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Executive Summary

Technological developments in artificial intelligence (AI), digitalisation, computational biology, and synthetic biology have fuelled a "21st-century biotechnology revolution". AI and innovative digital tools can play an important role in supporting the preparation and response to health emergencies. However, the use of these new technologies has also triggered concerns regarding their potential exploitation by individuals with ill intent. Therefore, relevant discussions and actions are needed to prevent their misuse.

This scientific background document explores the potential applications of AI in strengthening the EU biosecurity framework. The EU biosecurity framework comprises the policies, strategies, regulations, and operational measures designed to prevent, detect, and respond to biological risks.

The study is organised in three main chapters as follows:

The first chapter, *Introduction*, presents the study scope, an overview of the current state of the art and defines the three objectives of the study, which are:

1. Assess AI's potential to enhance the screening of synthetic nucleic acid orders, a countermeasure to prevent and detect access to synthetic nucleic acids with threat potential.
2. Identify and evaluate AI-based tools that enhance the detection and attribution of genetic engineering activities.
3. Assess both the opportunities and risks of AI in relation to Biosecurity in the EU and globally.

To meet these objectives, the study team conducted a **literature review using the academic database PubMed complemented by additional desk research** of relevant reports by experts in the field. Additionally, **key experts from academia, government and industry were interviewed** to collect further insights on the topic. In-depth interviews allowed the identification of the most critical research opportunities in the field while recognizing possible risks and obstacles to adopting AI for a reinforced EU biosecurity framework. The methodology is further detailed in the Annex.

The second chapter, *Results*, summarises the main outcomes of the study, which are organised into five sections.

Section 2.1 provides an initial overview of the opportunities that AI offers to enhance biosecurity. Some key highlights are AI's potential to process large datasets, perform advanced pattern recognition, automate tasks, facilitate access to knowledge, and contribute to the faster and more precise design of biological components. Since AI is regarded as a dual-use technology with both beneficial and potentially harmful applications, Section 2.3 discusses the risks associated with AI in biosecurity. The most significant risk identified is the potential for Large Language Models (LLMs) and Biological Design Tools (BDTs) to provide access to knowledge and capabilities for developing novel, more dangerous biological agents. This could aid malicious actors and complicate biosecurity efforts.

Section 2.3 presents seven tools used for screening orders of synthetic nucleic acid. Among these, four tools have integrated AI techniques, while the others use classical computational methods, often in the form of alignment algorithms. Evidence suggests that, currently, the main potential of AI in this context is augmenting the reference data sets used for screening with sequences that do not exist in nature, thus increasing the possibilities for detecting sequences of novel AI-generated proteins and reducing the occurrence of false positives.

Section 2.4 introduces the nine tools identified for their ability to detect genetic engineering and attribute it to the original author. Most of these tools utilise deep learning algorithms, stem from research initiatives, and are not currently on the market. Key recommendations to improve these biosecurity safeguards include integrating high-quality nucleic acid sequence datasets

to train deep learning models and adopting a combined approach of AI and other computational methods to enhance detection rates. The study highlights the potential of the tools but emphasises the need for further development to ensure real-world applicability.

Section 2.5 provides a final overview of the applications of AI for enhancing the EU biosecurity framework, from biosurveillance and early warning systems to outbreak monitoring and biothreat response and mitigation.

The thirds chapter, *Discussion & Conclusions,* reviews the key findings of the study and alignment with the three study objectives. First, it assesses AI's role in improving synthetic nucleic acid screening, highlighting its contribution to managing datasets, optimising sequence comparisons, and predicting biological entity structures. Second, the study identifies AI-based tools for detecting and attributing genetic engineering activity, revealing that AI techniques like deep learning show promise but require further development for real-world application. Third, the study evaluates the opportunities of AI technologies in biosecurity, noting that AI could enhance surveillance, early warning, outbreak monitoring, and response. The study also considers the risks, particularly concerning dual-use capabilities and the potential for malicious applications. Overall, the study underscores the need for policies to balance the benefits of AI-driven innovation with the management of emerging biosecurity risks.

Moreover, **this chapter proposes recommendations for policymakers to guide future actions in leveraging AI for an enhanced biosecurity framework in the EU**. These recommendations include the following:

- Promote multidisciplinary and cross-border collaboration in the fields of AI and biosecurity, to ensure a good alignment between the technical, ethical, and security implications.
- Support Research, Innovation and Capacity building in AI for Biosecurity to promote the development and adoption of AI tools that enhance pandemic preparedness and response
- Develope Standards and Guidelines for the use of AI and biotechnology in Life Sciences to promote the responsible use of technology against the misuse of biological systems and products
- Establish rules and legislation for AI in the life sciences to improve EU's Biosecurity.

The **study highlights the potential of AI in enhancing biosecurity, enabling the EU to detect and respond to emerging biological threats**. By leveraging AI, the EU can strengthen its biosecurity infrastructure. However, achieving this requires international and multidisciplinary collaboration, robust ethical guidelines and regulatory frameworks, and investment in research and innovation to ensure responsible and effective AI use. **As biosecurity challenges evolve and technological advancement progress, the EU's focus on innovation, cooperation, and adaptive governance will be crucial in safeguarding public health and global security**.

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Acronyms and abbreviations

AI	Artificial Intelligence
ANN	Artificial Neural Networks
BIS	Bureau of Industry and Security
CBRN	Chemical, Biological, Radiological and Nuclear materials and agents
CBB	Danish Centre for Biosecurity and Biopreparedness
DNA	deoxyribonucleic acid
EBRF	European Biosecurity Regulators Forum
EU	European Union
GEA	Genetic Engineering Attribution
GED	Genetic Engineering Detection
HERA	Health Emergency Preparedness and Response Authority
IBBIS	International Biosecurity and Biosafety Initiative for Science
IGSC	International Gene Synthesis Consortium
ML	Machine Learning
OFAC	Office of Foreign Assets Control
RF	Random Forest
RFID	Radio-frequency identification
RNA	ribonucleic acid
RNN	Recurrent Neural Networks
RPD	Regulated Pathogen Database
SD	System dynamic (simulation)
SDN	Specially Designated Nationals
SOC	Sequence of concern
WHO	World Health Organisation

Glossary

Term	Description
Artificial Intelligence system	Machine-based system that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments.
Biological threat	A biologic threat is an infectious disease with the potential to spread and cause an emergency. Infectious disease emergencies may occur naturally or be released unintentionally or intentionally to hurt people, create fear, and/or disrupt society.
Biosecurity	Policies, principles, technologies and practices implemented for the protection and control of and accountability for biological material, technology and information or the equipment, methods, skills and data related to their handling. Biosecurity aims to prevent intentional or accidental unauthorized access to, and loss, theft, misuse, diversion or release or even weaponization of such commodities.
Cybersecurity	Cybersecurity is the practice of protecting systems, networks, and programs from digital attacks.
Data integrity	Data integrity is the accuracy, completeness, and quality of data as it's maintained over time and across formats.
Dual-use items	Dual-use items are goods, software and technology that can be used for both civilian and military applications.
Gene synthesis	Gene synthesis is the artificial construction or manufacturing of nucleic acid sequences. This process involves creating new nucleic acid sequences in the laboratory using chemical or biological methods. The primary objective of this technology is to design and build nucleic acid sequences with specific functions or properties.
Genetic engineering	Genetic engineering (also called genetic modification) is a process that uses laboratory-based technologies to alter the DNA makeup of an organism.
Machine Learning	Branch of AI and computer science that focuses on using data and algorithms to enable AI to imitate the way that humans learn, gradually improving its accuracy.
Neural network	A neural network is a machine learning program, or model, that makes decisions in a manner similar to the human brain, by using processes that mimic the way biological neurons work together to identify phenomena, weigh options and arrive at conclusions.
Nucleic Acid	A chemical compound that serves as the main information-carrying molecule of the cell and that directs the process of protein synthesis, thereby determining the inherited characteristics of every living thing. The two main classes of nucleic acids are deoxyribonucleic acid (DNA) and ribonucleic acid (RNA).
Sequencing	Sequencing nucleic acids means determining the order of the four chemical building blocks - called "bases" - that make up the nucleic acid molecule.
Sequencing (Next-Generation sequencing)	Next-generation sequencing (NGS) is a massively parallel sequencing technology that offers ultra-high throughput, scalability, and speed. The technology is used to determine the order of nucleotides in entire genomes or targeted regions of DNA or RNA.
Synthetic biology	Synthetic biology is a field of science that involves redesigning organisms for useful purposes by engineering them to have new abilities.

1. Introduction

1.1. Study Scope: context and background

Technological developments in synthetic biology, Artificial Intelligence (AI), digitalisation, automation, and computational biology are fuelling a "21st century biotechnology revolution" with lower costs and easier access to the synthesis and modification of biological material (Carter et al., 2023a). The speed at which these technologies are evolving often exceeds the capacity of government regulatory oversight, leading to possible gaps in policies and regulations (Mouton, Lucas, & Guest, 2023). While this revolution opens the doors to scientific discoveries and innovation (i.e., supporting the modernisation of research within the health sector), it also triggers concerns with respect to the possibility of individuals with ill intent to exploit them for destructive purposes.

In the aftermath of COVID-19, the Health Emergency Preparedness and Response Authority (HERA) was set up to anticipate health threats and crises. It collaborates with EU Member States to enhance preparedness and coordinates efforts in response to health crises (Directorate-General for Health Emergency Preparedness and Response Authority, n.d.). HERA is bringing forward key initiatives to promote advanced research and development of technologies for tackling priority cross-border health threats (European Commission, 2023). In parallel, the European Commission is leading a coordinated effort to embrace AI solutions that aim to address public health and global crises (Fontes, Corrigan, & Lütge, 2023).

When the connection between opportunities and potential risks arising from this biotechnology revolution became evident, **the concept of biosecurity emerged to safeguard society from the unintended or malicious consequences of biotechnological advancements**. Different definitions of biosecurity have been adopted over the years. In Denmark, the Danish Centre for Biosecurity and Biopreparedness (CBB) defined biosecurity as "*Measures designed to prevent the malicious abuse of "dual use" biological materials, i.e. biological substances and related materials that are intended for legal research, but which could also be used to create and disseminate a danical agent*" (Centre for Biosecurity and Biopreparedness (CBB) Denmark, 2015). In this definition, biosecurity is limited to the prevention of intentional misuse of "dual use" biological materials. This is different from biosafety definition which also covers the prevention of accidents or unintentional misuse. This report adopts the **definition of biosecurity outlined in the World Health Organisation (WHO) 2024 laboratory biosecurity guidance.** This definition has a broader scope, covering both the intentional and accidental improper or harmful use of biological material and related resources (See **Error! Reference source not found.**).

This study will explore the use of AI to enhance biosecurity measures supporting HERA's mission of preparing and responding to health emergencies. For this, this report uses the **updated OECD definition of 'AI systems'** to ensure technical accuracy and international alignment (See **Error! Reference source not found.**).

Box 1 *Working definition of "Biosecurity" and "AI system" in this report*

Definition of Biosecurity by the WHO 2024 Laboratory biosecurity guidance.

"Policies, principles, technologies and practices implemented for the protection and control of and acountability for biological material, technology and information or the equipment, methods, skills and data related to their handling. Biosecurity aims to prevent intentional or accidental unauthorized access to, and loss, theft, misuse, diversion or release or even weaponization of such commodities." (World Health Organization, 2024)

Definition of AI system by OECD.

"An AI system is a machine-based system that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments. Different AI systems vary in their levels of autonomy and adaptiveness after deployment."

1.2. Study objectives

This scientific background **evaluates the potential applications of AI in enhancing biosecurity in the EU**. Through a comprehensive literature review, desk research and consultations with key experts, the study maps existing technologies and solutions aimed at improving biosecurity. It seeks to identify key research opportunities and findings in this field while also addressing potential risks and challenges associated with adopting AI to strengthen biosecurity in the EU.

More specifically, the report has the following key objectives:

1. Assess AI's potential to enhance **screening of synthetic nucleic acid orders**, a countermeasure to prevent access to bioweapon components
2. Identify and evaluate AI-based tools and methodologies that enhance the **detection of unauthorised genetic engineering activities, and improve attribution capabilities** to trace the origin of genetic modifications
3. **Evaluate the risks posed by AI technologies and the opportunities they offer to enhance biosecurity** both within the EU and globally.

The report will support HERA's efforts in investigating how AI tools can be leveraged for a strengthened biosecurity in the European Union for an effective policy development. To do this, the report will outline the needed targeted actions to adopt AI-supported technologies in the field of biosecurity and further explain how EU crisis preparedness and response would benefit from the adoption of such AI-based tools.

1.3. Structure of the report

The study is divided into three main chapters and and annex:

Chapter 1 – Introduction presents the purpose of the study together with the background and overview of the state of the art concerning biosecurity in the EU.

Chapter 2 – Results presents the key findings that emerged from the literature review, according to the above-mentioned study objectives. These are organised into three main parts:

- AI's potential for an enhanced screening of synthetic nucleic acid orders
- AI-enabled tools to improve the detection and attribution of genetic engineering
- AI in the overall EU biosecurity framework, including opportunities and risks posed by AI to biosecurity

At the end of each results section, key insights resulting from the consultation activities with experts in the field are presented to complement the outcomes of the literature review.

Chapter 3 – Discussion and Conclusions includes final considerations based on the analysis of the key findings of the study, including identified gaps and possible directions for future research and policy actions.

Annex – Methodology outlines the methodology followed in this report, detailing the key research questions, the literature search strategy, and the data extraction methodology. A PRISMA diagram is presented to give a clear snapshot of the information considered in the scientific review.

1.4. Defining AI: Context and Techniques

AI is a vast field of study that includes a variety of methodologies and applications, such as computer vision, natural language processing, and machine learning for pattern recognition, prediction, and large-scale data analysis. In 2023, member countries of the Organisation for Economic Co-operation and Development (OECD) approved a revised version of the

organisation's definition of an AI system, which was also used as a relevant basis for the EU AI Act (OECD, 2023).

This study uses the updated OECD definition of 'AI systems' to ensure technical accuracy and international alignment. The definition states that "An AI system is a machine-based system that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments. Different AI systems vary in their levels of autonomy and adaptiveness after deployment."

Moreover, the OECD developed the Framework for the Classification of AI Systems to evaluate and characterise AI systems across key dimensions such as model type, data and input, main tasks, and examples (OECD, 2022). Using this framework, **a classification model for this study was developed to assess and categorise AI systems based on their characteristics**. This model identifies three main types of AI Systems in this study: Symbolic AI models, Statistical AI models, and Hybrid AI models. Table 1 provides an overview of the three categories considered, highlighting their main characteristics and tasks. **Symbolic AI models** (or knowledge-based models) are used for tasks such as planning and optimisation, allowing inferences and classification activities. **Statistical AI models** are used for trend and pattern prediction based on data (e.g., linear regression, logistical regression, decision trees, neural networks, naïve Bayes, deep learning). Lastly, **Hybrid AI models** combine the previous approaches to extract complex features, enhance pattern recognition, and learning abilities (e.g., natural language processing (NLP) algorithms).

Table 1 *Definition of the AI techniques considered in this study*

	Symbolic AI model	Statistical AI model	Hybrid AI model
Description	Symbolic or **knowledge-based** AI uses human-generated logical representations to infer a conclusion from a set of constraints (variables). These constraints include **rules**, **ontologies** and **search algorithms** that rely on explicit descriptions of variables and descriptions of the interrelations between these variables.	**Statistical AI models** (e.g. genetic algorithms, neural networks and deep learning) identify patterns based on data rather than expert human knowledge. Primarily used for recognition, more recently, these models are also being used for tasks such as synthesising and generating images or audio.	Many applied AI systems **combine symbolic and statistical** models into "hybrid" models. For example, NLP algorithms often combine statistical approaches building on large amounts of data and symbolic approaches that consider issues such as grammar rules.
Examples	Knowledge-based and expert systems	Neural networks, Deep learning, Linear regression, Logistical regression, Decision trees	Natural Language Processing
Data	Labeled and categorised	Unlabeled and uncategorised	Can be labeled or unlabeled
Task	Making inferences and performing classification	Trend/ pattern prediction	Complex feature extraction and pattern recognition/ learning abilities

Source: *author's elaboration based on literature review findings.*

As AI systems become more advanced and diverse, their application extends beyond traditional fields into areas like synthetic biology and biotechnology. These technologies rely on AI models to design, predict, and optimise biological processes, enhancing both research capabilities and practical applications. The convergence of AI and biology is particularly significant at the 'digital-physical interface,' where an actor integrates digital design and outputs into a biological reality (Carter et al., 2023a). **The interface between physical and digital elements is considered by experts as a crucial juncture for enhancing biosecurity** (O'Brien & Nelson, 2022).

1.5. Overview of existing safeguard measures

Advancements in biotechnology and artificial intelligence, while offering new possibilities for medical and agricultural innovation, also raise concerns about potential misuse. The ability to design and manipulate biological systems digitally introduces new vulnerabilities, particularly as the tools for genetic editing and synthesis become increasingly accessible. Consequently, **ensuring that these powerful technologies are used responsibly has become a priority for policymakers and scientific communities alike.** Efforts to strengthen biosecurity and manage biorisks have increased globally and within the EU, leading authorities to implement measures to control sensitive biological materials.

The growing demand in the gene synthesis market (Global Market Insights, Inc., 2022) makes the establishment of safeguards for nucleic acid synthesis technology essential. Robust safeguards and control can prevent the unintentional distribution of biological components that could be used to create harmful biological agents. **Current biosecurity safeguard measures involve screening synthetic nucleic acid orders** to assess the safety and legality of requested sequences and verifying customer information. The goal is to prevent misuse while allowing scientific and medical research to continue safely.

Most governments, including those in the EU, do not mandate screening practices by law (Hoffmann, et al., 2023). Various initiatives have been launched to address this regulatory gap. In 2009, the International Gene Synthesis Consortium (IGSC) created a screening protocol to set standards and best practices aimed at preventing the misuse of nucleic acid synthesis (International Gene Synthesis Consortium, 2017). The WHO, in 2022, recognised the potential risks related to nucleic acid synthesis and recommended adherence to established screening protocols (World Health Organization, 2022). At the national level, especially in the US and UK, Governments have published national screening guidance for users and providers of synthetic nucleic acids (UK Government, 2024). In addition to voluntary efforts and adherence to common guidelines, some countries, such as the US, introduced mandatory measures for screening. In 2024, the White House's Office of Science and Technology Policy introduced the Framework for Nucleic Acid Synthesis Screening, directing providers to screen orders for sequences of concern (SOCs) and verify customer legitimacy, making it mandatory for federally funded research (The White House, 2024) (Diggans & Leproust, 2019).

The screening of synthetic nucleic acid orders largely depends on the existence of repositories of well-characterised regulated pathogens. A shift towards a less centralised nucleic acid synthesis market and the rapid developments in synthetic biology could lead to the evasion of current screening mechanisms. Experts emphasise the need for alternative safeguard measures that go beyond list-based sequence screening (Batalis, Schuerger, Gronvall, & Walsh, 2024).

A biosecurity safeguard measure gaining attention is the detection of genetic engineering (GED). This process identifies whether an organism's genetic information has been artificially modified. It helps detect unauthorized or harmful genetic modifications, allowing authorities to identify potential misuse, such as engineered pathogens. GED can be **complemented by genetic engineering attribution (GEA), which traces the source of genetic modifications**. GEA aids in investigations to identify the parties responsible for such modifications. These safeguard measures aim to directly identify genetic engineering activity and trace it back to its source, making biosecurity measures more independent of reference databases of well-known pathogens.

Biosecurity safeguard measures demonstrate ongoing efforts to manage biosecurity risks at the operational level. However, their development and implementation are rooted in a **broader international policy framework designed to prevent the misuse of biological materials and technologies**.

The key regulatory instruments for stronger global biosecurity include the Biological Weapons Convention (BWC)[1], the United Nations Security Council Resolution 1540 (UNSCR 1540)[2], and the International Health Regulations (IHR)[3]. These three instruments, adopted by the 27 EU Member States, **are the only three legally binding international instruments forming the backbone of the global effort to prevent the development of biological weapons and enhancing global biosecurity** (Vennis, Boskovic, Bleijs, & Rutjes, 2022) (United Nations Office for Disarmament Affairs., 2024). **At the EU level, several policies have evolved in parallel to improve biosecurity**. To support the BWC and the implementation of the UNSCR 1540, the European Council adopted the European Security Strategy[4] and the European Strategy against the Proliferation of Weapons of Mass Destruction[5] along with several Joint Actions and Council decisions (Wakefield, 2022). In 2009, the European Commission adopted the CBRN Action Plan to address threats from chemical, biological, radiological, and nuclear incidents of accidental, natural or intentional origin (Bielecka & Mohammadi, 2014). More recently, the COVID-19 pandemic highlighted vulnerabilities, triggering investment in biosecurity projects that focus on the development of medical countermeasures for CBRN threats.

Technological innovations are transforming the threat landscape, potentially shifting attention from state-sponsored actions to micro-level risks posed by smaller entities or individuals (Lee, Cowan, & & Tankard, 2022). **Despite the robust regulatory frameworks, biosecurity remains a complex concept across the EU Member States**. Different national priorities, resources, capacities, and existing legal frameworks lead to varying approaches to managing biological risks. As a result, **biosecurity in the EU cannot be viewed as a singular entity but rather as a heterogeneous framework**, consisting of a system of policies, strategies, regulations and operational measures to prevent, detect, and respond to biological threats that reflect the different approaches and capacities of individual Member States.

Moreover, as the interface between digital and biological systems continues to grow, it is essential to **evaluate the evolving role of AI and innovative digital tools in strengthening biosecurity measures, assess the new challenges, and adapt future biosecurity policies** to mitigate biological risks effectively.

2. Results

This section summarises the main results of the scoping literature review. The first part focuses on how AI can contribute to the overall EU biosecurity framework and the associated risks it entails. This section is followed by a presentation of the identified tools for screening synthetic nucleic acid orders and how AI has been incorporated to enhance their use, along with a discussion on the application of AI to improve genetic engineering detection and attribution. The results section concludes with a general outlook and presentation of potential future applications beyond the screening of orders, GED and GEA. Additional references are available at the end of each subsection, based on information gathered through in-depth structured interviews with key experts in this field.

[1] (United Nations Office for Disarmament Affairs., 1972)
[2] (United Nations Office for Disarmament Affairs, 2004)
[3] (World Health Organization, 2016)
[4] (European Council, 2003)
[5] (Council of the European Union, 2003)

2.1. AI's potential in the overall EU biosecurity framework

AI's capacity to integrate large datasets, quickly analyse complex patterns, and deliver real-time risk assessments can be applied in the detection, analysis and response to both natural and deliberate scenarios of biological threats. These advanced capabilities are crucial for strengthening the EU's biosecurity framework, as they allow for earlier detection, quicker responses, and more effective threat mitigation strategies.

One of the basic areas explored is the use of **AI to automate and accelerate certain steps in scientific discovery** like generating hypotheses, literature screening, assisting in the design of experiments and data collection exercises, automating code writing and programming, etc. Although promising, the limited knowledge of common users could lead to an overestimation of AI-based automation (Carter et al., 2023a) (InterAcademy Partnership, 2024).

The literature review revealed **Machine Learning use cases for enhancing pandemic preparedness and response**. Syrowatka et al. (2021) identified the potential of AI in forecasting infectious disease dynamics, surveillance and detection of outbreaks, monitoring adherence to public health recommendations, detection of symptomatology (e.g., body temperature), timely diagnosis of infection, assistance in patient triage, and prognosis assessment and patient's treatment response. Nevertheless, most of the identified uses were at a developmental stage, not ready to inform public health decisions.

Other publications focus on **how Large Language Models (LLM) and Generative AI (Gen AI) could be applied to biosecurity**. Some experts discuss that current general-purpose LLMs applied to the life sciences are unlikely to provide significant novel insights into biological systems and therefore are only useful for rather unexperienced individuals with a shallow knowledge on the topic of interest. However, experts argue that this trend could easily change, considering the rapid advancement of this technology (Carter et al., 2023a). Chen et al. (2024) reviewed the potential role of **LLMs for automating biosecurity tasks,** which included data gathering from public data sources (e.g., papers or news), report summarising, and reviewing safety forms. The authors concluded that while LLMs have great potential, they are limited by their provision of shallow or incorrect information. Nevertheless, through an in-depth analysis of 169 publications on AI-enabled synthetic biology, Undheim (2024) identified that **AI-based tools are already being used for several standard and operational functions in synthetic biology** like text classifiers, generic search engines, generation of ideas, helping with access to scientific knowledge, proofreading, summarizing knowledge, translation, text mining of biomedical data, optimizing coding and workflow optimization among other similar functions.

Other authors have focused on **the use that AI-based biodesign tools (BDT) can have for biosecurity**. AI-based BDTs are AI systems trained on biological data that can help design proteins or other biological agents. Different from the general-purpose LLMs, AI-based BDTs are reserved for those with a deeper scientific experience and knowledge. BDTs, such as RFDiffusion and protein language models like ProGen2 and Ankh, can expand the ability of experienced actors to use available information to design relevant biological components (Sandbrink, 2023). The **capabilities of BDTs to facilitate the design, testing and optimization of new biological components with increased accuracy has many beneficial applications** in fields like medicine, the pharmaceutical industry and agriculture. In these areas an improved design of vaccines and novel therapeutics could be extremely beneficial (Carter et al., 2023a). For instance, AlphaFold2 is an AI BDT currently used in the field. Although it doesn't directly generate biological designs, it features an AI-enhanced ability to predict protein 3D structure (Senior, et al., 2020). Despite the promising advances, there are still some challenges to overcome in this field like the transition of structure-based biodesign to a function-based one or the capacity to design those biological components which are less well characterised and for which little data is available.

2.1.1. Considerations from the experts

The findings above present the information gathered from reviewing the literature. These results have been further discussed and validated with experts in the field, who provided an overview of the role of AI in biosecurity. Box 2 presents the main learnings from the interviews conducted with several experts from Academia, Industry, Government, and International Agencies.

Box 2 Considerations from the experts - AI in the overall EU biosecurity framework

Highlights resulting from Experts' consultation

Opportunities for the use of AI

Several policy and industry experts pointed to the opportunities of AI to improve biosecurity while promoting innovation: AI for medical countermeasure development; AI for bio surveillance; AI for broader forensic capabilities (e.g. monitoring and screening cloud laboratories to avoid cyber-attacks); and AI for collecting data, such as open-source information to better understand the biological threat landscape. Several interviewees identified challenges to overcome to maximise the use of AI for biosecurity:

- AI experts not being interested in biosecurity matters and biosecurity experts lacking expertise on AI. **Disaggregated knowledge** in this field slows down concerted efforts among different actors.
- One academic expert underscored that, due to the novelty and misinformation around AI, several **scientists are still reticent to use AI** in their research. This is lost knowledge and potentially innovation because the tools are often developed by computational dry laboratories and don't necessarily get adopted into wet labs.
- In a similar way **disaggregated data** is a challenge for using AI to do any kind of bio surveillance. With respect to the opportunities of AI, one academic expert underlined that if **data is stored in one spot**, there is a lot of opportunity for monitoring human health. If it is disaggregated, then it would **limit the opportunities that AI can provide**, e.g. from electronic health records. Another important aspect related to data is the **integration of disparate data types** to make sense of things in aggregate form, e.g. human movement or migration, animal population density or weather conditions. Questions that rely on deriving insight from vast amount of data are where AI can help.
- **Low commercial incentives** in the biosecurity field. A synthetic nucleic acid screening provider started seeing decreasing similarity to naturally occurring sequences from the orders received, due to AI design tools. In this context, they were advocating for government to invest in research programs to solve this community challenge.
- Most stakeholders underlined the need for **increased collaboration among industry, academia and government**. Several experts underlined that government is critical to enforce rules but does not have the capacity alone to set standards because it doesn't have the knowledge that industry and academia can bring to the table. Academia has a critical role to play in answering the question "What should we control? What is the actual risk level associated with different genes?" Academia is also a great place for thinking about potential types of evasion that we need to defend against, including AI based evasion. Industry has pragmatic experience as it must deal with false positive rates and the cost of decision making. Thus, it should be at the table concerning these aspects because it must deal with the actual implementation and solutions.
- Finally, several experts underlined that, taking the example of the US Executive Order, **Regulations** could push things ahead. Government experts highlighted that they are still waiting for an EU directive on Biosecurity to be implemented in Member States. If countries don't have a law, then it's also difficult to find a contact person, which in turn also affects collaboration with other stakeholders in this field.

2.2. The potential risks posed by AI to biosecurity

As AI advancements fuel innovation and technological developments, they can also enable their misuse as biological weapons (de Lima, et al., 2024). Because of the consideration of AI as dual-use technology, it is relevant to consider the concerns shared by the scientific community related to the risks that AI might pose for global biosecurity.

The InterAcademy Partnership (IAP) meeting in Trieste brought together technical and scientific experts from different countries to discuss the potential impact of AI on global biosecurity and international cooperation within the framework of the BWC. Some participants warned that **AI could negatively impact biosecurity by enabling the faster, cheaper creation of biological weapons, including AI-designed toxins or contagious proteins like prions** (InterAcademy Partnership, 2024).

Sandbrink (2023) highlights two classes of AI tools that could pose biosecurity risks: Large Language Models (LLMs) and Biological Design Tools (BDTs). In a recent article, Batalis and colleagues (2024) also argue that LLM-based chatbots and BDTs could influence biosecurity differently. While **LLM-based chatbots lower barriers for non-expert individuals to access information** that could be misused for malicious intent, **BDTs can provide new capabilities to users with ill intents**.

Empirical evaluations suggest that LLMs may aid in planning biological attacks to some extent (Soice et al., 2023), though other studies report no effect (Mouton, Lucas, & Guest, 2023) (National Security Commission on Emerging Biotechnology (NSCEB), 2024). Some experts warn that LLMs could facilitate familiarising individuals with biological agents and resources needed to develop harmful agents (Carter et al., 2023a). However, no documented cases of AI-enabled bioweapons exist, and experts note that significant hurdles remain, including access to biological components, laboratory infrastructure, and specialised training to test and deploy the designed agent. Expert opinions differ on how much tacit knowledge LLMs can effectively provide and their actual impact, given the potential inaccuracies of the model (Carter et al., 2023a).

Regarding the risks associated to BDTs, Carter and colleagues (2023a) noted that their misuse often stems from their original benign purpose, such as predicting high-pandemic-potential pathogens, which could also identify promising candidates for biological weapons. Experts interviewed by Carter in their study believe that **AI BDTs have a higher possibility than LLMs to generate biological designs for toxins, pathogens, or other biological agents** that may be more harmful than those found in nature. Moreover, the design of new types of biological agents could raise the ceiling of potential harms and **evade synthetic nucleic acid screening by redesigning known hazards**.

Sandbrink (2023) declares that the combination of both tools could significantly expand the capability and accessibility to manipulate biological agents, increasing the risks of nefarious actions. Nevertheless, he emphasises the **need for evidence to confirm whether these tools actually lower barriers to biological weapons creation**. Some research groups are actively researching the risks of AI-assisted bioterrorism including the use of LLMs, but model evaluations have limitations as a means for ensuring safety.

Experts highlight that **AI tools can also be used to spread targeted disinformation that may erroneously influence decision-making around response to biological threats** or compliance with the BWC obligations and international collaboration. In terms of biosecurity and collaboration, experts also noted that **AI tools could also be used to identify gaps and vulnerabilities in healthcare and threat response systems** that can be exploited by adversaries.

The literature also explores the mitigation measures that could be put in place to reduce the identified risks of AI for biosecurity, like the release of AI models tested for safety. However, some authors claim that, even though responsible AI laboratories may release biologically

capable AI models only when they have been trained to be safe, skilled actors could overcome these guardrails through jailbreaks or other means (Mazeika, et al., 2024) (Qi, et al., 2023). Moreover, other authors underlined that the availability of open-source codes could render even capable AI models subject to malicious actors who can remove such safeguards (Gopal, et al., 2023). Finally, experts recall that AI tools produce digital outputs rather than physical ones. Creating a biological agent or substance (including biological weapons) requires access to laboratories, production equipment, and technically proficient personnel which is difficult to obtain.

Table 2 presents some of the key insights from the literature on the risks that AI-enabled tools pose to the advancement of biotechnology, synthetic biology and biosecurity, especially considering a non-state actor perspective.

Table 2 Key risks posed by AI for biosecurity

Risk	Current safeguards that prevent the risk from materialising	Potential additional mitigation strategies	Source
Chatbots may facilitate knowledge access to create biological threats. Biological Design Tools (BDTs) may provide additional capabilities to users with malicious intents	• Availability of the appropriate infrastructure (laboratory, reagents, etc.) • Practical knowledge of how to work in a laboratory • Access to funding	• Mandatory synthetic nucleic acid order screening • AI model evaluation (pre and post release) with independent audits	O'Brien and Nelson (2020) Carter et al (2023a) Batalis et al (2024) De Haro (2024)
BDTs may help generate toxins or pathogens that are not well characterised in the scientific literature, making it more difficult for nucleic acid providers to conduct effective screening of synthesis orders.	• Resources required for testing BDT's candidate designs (time, funding, expertise, etc.) • Controlled access to AI models (authentication) and to the infrastructure needed to train them	• Improvement of screening tools to keep pace with biodesign development • Include protein structure predictor in screening methods • Train AI models to refuse to engage in specific topics (i.e. "red-teaming" to assess misuse potential)	O'Brien and Nelson (2020) Carter et al (2023a) De Haro (2024)

Source: *author's elaboration based on literature review findings.*

2.2.1. Considerations from the experts

The findings above present the information gathered from reviewing the literature. These results have been further discussed and validated with experts in the field, who provided an overview of the risks of AI for biosecurity. Box 3 presents the main learnings from the interviews conducted with several experts belonging to Academia, Industry, Government, and International Agencies.

Box 3 Considerations from the experts í The potential risks posed by AI to biosecurity

Highlights resulting from Experts' consultation

Experts coming from industry, academia and government also pointed to the inherent risks of using this technology within the biosecurity field. Indeed, **AI-enabled biological misuse** was considered by several experts as a huge risk, i.e. the potential of AI models being used to come up with new sequences of concerns that might cause harm. One academic and industry expert recognised the need to ensure that future models do not suggest ways of building nastier things than nature does. LLMs that are as good as any living biotech researcher at thinking about new designs and better at considering security matters can be a major concern.

In addition to this, several experts pointed to the risk of **AI allowing to troubleshoot tasks**. According to them, tacit knowledge barriers, i.e. having pragmatic knowledge of working in a laboratory, have contributed to making the development of deadly pathogens harder but this is changing with AI. Being able to evaluate the level of capabilities that AI models offer or how much uplift they're able to offer actors is an important part of being able to monitor for AI risks. According to one policy expert, in the short term, the risks may outweigh the benefits. Then in the longer term, AI could be the biggest asset to security.

Another major risk related to the use of AI for intelligence gathering is **misinformation**, both in terms of diluting important information, as well as creating false data.

Academic and policy experts also mentioned relevant **ethical considerations and risks** of integrating AI in this field, especially when using clinical data. There are important data privacy concerns on giving access to automated algorithms to databases with health information.

2.3. AI-enabled tools for an enhanced screening of synthetic nucleic acid orders

These are the main takeaways on the role of AI for enhancing the screening of synthetic nucleic acid orders.

- The study identified seven tools being used or under development for the screening of synthetic nucleic acid orders. Four of the identified tools have some AI techniques integrated within their operating methods or databases. The other three use classical computational methods often in the form of alignment algorithms.
- Given the voluntary nature of screening synthetic nucleic acid orders, providers are not incentivised to improve their current screening methods integrating AI. However, experts consider that this may rapidly change considering the new US legislations to strengthen biosecurity (see section **Error! Reference source not found.**).
- Looking at the future, AI has the potential to enhance the screening of synthetic nucleic acid orders in two ways:. i) augmenting reference data sets with sequences that do not exist in nature and thus increasing the possibilities to screen also for sequences of novel AI-generated proteins and ii) enabling real-time structure prediction, which is currently computationally resource intensive.

The general lack of specific requirements that mandate the screening of synthetic nucleic acid orders, leaves providers to voluntarily implement screening procedures based on high-level protocols and recommendations. A recent study by Kane and Parker (2024), which interviewed providers and associated stakeholders from 18 organisations, highlights significant variability in screening practices and a general lack of regulation and effective governance.

Many providers either do not screen synthetic oligonucleotides or lack procedures for reporting suspicious orders to law enforcement. The study found that the **most common tool used for screening of potential orders** against SOCs was the US National Centre for Biotechnology Information's (NCBI) **Basic Local Alignment Search Tool** (BLAST) integrated into automated algorithms. BLAST searches for regions of significant similarity between nucleotide or peptide sequences and databases. With many biological applications, it can also aid providers in screening orders (Altschul, Gish, Miller, Myers, & Lipman, 1990).

Besides this basic bioinformatic tool, the study team identified six advanced tools available on the market for screening synthetic nucleic acid orders. The tools are described below. This

section presents the outcomes of the scoping literature review focused on the potential use of AI to improve synthetic nucleic acid orders' screening tools or processess.

- **Aclid Platform** performs customer screening and alignment-based sequence screening integrating automated and AI-driven data curation (Wheeler, et al., 2024a). The Aclid Platform uses protein language models like ESM to generate embeddings to search for similar or homologous proteins in vector space and for predicting structures to perform structural alignment. The outputs from the searches and predictions are used to annotate their database, improving their false-positive and false-negative rates.
- **Common Mechanism for DNA synthesis screening** (Beta version), an open-source, free, globally available tool for DNA synthesis screening to help providers of synthetic DNA and RNA effectively screen orders. It provides automated software for screening sequences of nucleic acids and resources like guidelines to facilitate customer screening. It is intended as a common standard and baseline for screening providers worldwide (Wheeler, et al., 2024b).
- **Fast-NA Scanner,** a proprietary software designed to integrate with workflows for screening nucleic acid sequences of more than 50 base pairs, delivering results about 1,000 times faster than BLAST. It uses Bloom filters, a probabilistic data structure allowing some false positive but no false negatives, to detect k-mer[6] signatures unique to specific pathogen or toxins. While it does not handle customer screening, the company recommends using established customer screening systems from the finance industry (Beal et al., 2019) (Beal et al., 2022) (Beal et al., 2023).
- **SecureDNA,** a free, non-profit screening platform designed to safeguard DNA synthesis while protecting customer data. It uses a privacy-preserving cryptographic algorithm to check short DNA subsequences from the orders (20 aminoacids) against a database of known hazards or predicted functional variants of controlled genes. Its database comprises hashed k-mers from conserved regions of SOCs and mutated variants that maintain biological function, enabling SOC inclusion without revealing their identity. The pricacy-preserving mechanism can protect trade secrets and shield research projects (Baum, et al., 2024).
- **SeqScreen** is a tool that beyond the taxonomic identification of known pathogens, it also reports functional threat assessments for both known and emerging pathogens. It feeds local alignments into a machine learning classifier trained to predict functional categories (ie., "confers antibiotic resistance") providing insights into why a particular sequence was identified as concerning (Alexanian & Langenkamp, 2024). Machine Learning techniques are being considered to further automate a portion of this work by predicting a suite of biological processes of interest on large protein databases before incorporation into the SeqScreen framework (Hoffmann, et al., 2023).
- **UltraSEQ** uses an information-driven alignment approach combined with machine learning to identify taxonomic best matches and SOCs. Advanced algorithms are used to facilitate organism and function identification. The associated metadata is the result of years of government-funded curation efforts, reflecting the significant investment required to develop a comprehensive set of annotated sequence records (Gemler, et al., 2023).

A pilot project aiming to develop a test set for evaluating the efficiency of screening tools showed that results varied significantly for certain genetic sequences (i.e. bacterial and fungal). This heterogenity reflecting the **variability and lack of uniformity among screening practices** (Wheeler, et al., 2024a).

[6] k-mer lookup is a bioinformatics approach which breaks up sequences into small sub-sequences of length k (èk-mersê), and then flags exact matches between the k-mers and sequences in a given database.

Table 3**Error! Reference source not found.**, below, provides a comparison of the current available tools to perform screening of synthetic nucleic acid orders. Given the proprietary nature of such tools and the information hazards linked to exposing specific techniques applied to screening, **in-depth information on the AI algorithms and systems integrated in such tools is often difficult to obtain**. The table, based on the publicly available data enriched through consultation activities with experts in the field, provides and overview of how computational/AI techniques are currently integrated in these tools and processes.

2.3.1. Challenges of AI used for synthetic nucleic acid order screening

Current screening solutions still require manual review of suspicious sequences, which can be complex to interpret and thus increase costs for providers (Diggans & Leproust, 2019). However, synthetic nucleic acid screening companies have started integrating AI algorithms into their screening methods in preparation to the development of the sector and new regulatory mechanisms to enhance biosecurity.

Moreover, Lee and colleagues (2022) have developed a **conceptual framework based on AI, the Biosecurity AI Network (BAIN),** that proposes to enhance biosecurity by integrating AI into current screening procedures of synthetic nucleic acid and peptide orders against a database with genomic and proteomic data of known pathogens, toxic peptides and proteins. The system could use AI and machine learning for predicting sequences' biological function "in silico" and to compile customer profiles, orders and screening outcomes to build a network map with user-submitted data and open sources to help detect anomalies. **BAIN aims to leverage AI to provide a more robust approach to biosecurity, addressing the evolving challenges posed by advances in synthetic biology and genetic engineering**.

Besides the promise of AI to enhance synthetic nucleic acid orders' screening, the literature identified four main challenges for the integration of AI for synthetic nucleic acid order screening. The main challenges are described below:

- **Scientific uncertainties.** The unknown role of certain sequences in pathogenicity, combined with the evolving nature of biological systems, means that any set of concerning sequences must be regularly updated and therefore AI systems might require re-training (Wheeler, et al., 2024b).
- **Complex sequence categorisation**. Scientific discoveries about pathogenicity sequences are often poorly integrated into databases. This, coupled with the heterogeneity on screening methods and various computational approaches leads to inconsistencies in identifying potentially hazardous sequences, making it challenging for manufacturers to make decisions based on such inconsistent and sometimes untraceable information (Kane & Parker, 2024).
- **Quality control of training databases.** The effectiveness of AI models relies on the quality of the training data. Predictions may be inaccurate or biased if the training data is skewed or of low quality. Given that data consistency and reliability can vary across sources, ensuring high-quality training data is essential for accurate AI-enhanced screening (Diggans & Leproust, 2019).
- **Technical challenges and increasing costs for providers**. The newly proposed HHS guidance has expanded recommendations for screening to shorter sequences (20 base pairs for high-mass orders (>1 mM)). This poses a new technical challenge and increasing costs, as the probability of false similarity between orders and SOCs increases as sequence length decreases (Hoffmann, et al., 2023).

Table *3* Commercially available tools for the screening of synthetic nucleic acid orders.

Name	Computational/ AI techniques	Data analysed	Accuracy	Target	Availability	Source
Aclid platform	AI-driven data curation. Use of protein language models (i.e., ESM)	Nucleic acid & protein sequences and customer screening	0.2-0.5% false-positive rates. Potential to go down as low as 0.05%.	DNA screening	On the market by Aclid	(Wheeler, et al., 2024a)
BLAST	Heuristic algorithm estimating best matches for a query sequence against a sequence DB	NCBI nucleic acid and protein sequences	N.A.	Sequence alignment (not designed for biosecurity purposesonly)	For research [publicly available]	(Altschul, Gish, Miller, Myers, & Lipman, 1990)
Common Mechanism for DNA synthesis screening	Compares sequences against a 'biorisk' database equipped with ML models (HMM[7]) capturing profiles of SoC	Publicly available data; standard protein and nucleotide DB	<2% false positive flags of real synthetic biology designs	DNA screening	For research and industry providers [publicly available]	(Wheeler, et al., 2024b)
Fast-NA Scanner	Based on the Framework for Autogenerated Signature Technology originally developed for the detection of malware in network traffic	Nucleic acid sequence data	<2% false positive rates	DNA screening	On the market by RTX BBN Technologies	(Beal et al., 2019) (Beal et al., 2022) (Beal et al., 2023)
SecureDNA	Shamir's secret sharing (SSS) algorithm	nucleic acid and peptide subsequence	Negligible false alarm rate (reverse screening system that removes known non-hazards from the database)	Cryptographic DNA screening; compatible with benchtop synthesizers	For research and industry providers [publicly available]	(Baum, et al., 2024)
SeqScreen	ML based classification: techniques: L1-regularization (Support Vector) and dropout (Neural Networks) to balance weights and reduce overfitting	Both nucleic and amino acid sequences	N.A.	Not explicitly designed for identifying biosecurity threats	For research and industry providers [publicly available]	(Balaji, et al., 2022)
UltraSEQ	Information-theory-based approach combined with ML algorithms	nucleic acid screening and dataset analysis (metagenomics)	N.A.	DNA and metagenomic screening	On the market by Battelle	(Gemler, et al., 2023)

Source: *author's elaboration based on literature review findings.*

[7] Hidden Markov Models, a statistical model that represents systems with unobserved states, where each state generates observable data based on specific probabilities. It is commonly used for sequence prediction, such as in bioinformatics.

2.3.2. Considerations from the experts

The findings in Section 2.3 were significantly complemented through the consultation activities with key experts, which allowed to validate the identified AI solutions and tools for the screening of synthetic nucleic acid orders. The main input from interviewed experts is summarised in Box 4.

Box 4 Considerations from the experts: AI-based tools for an enhanced screening of synthetic nucleic acid orders

Highlights resulting from Experts' consultation

Several industry, academic and policy experts working in this field highlighted that there are opportunities for **AI to make the screening of synthetic nucleic acid orders better** but currently it's not clear how challenging this is going to be.

AI's potential for enhanced screening

One government expert in the EU underlined that AI is going to be both an enabler and **the only way that we can screen orders** in the future. Another expert coming from academia and working for a nonprofit organisation on biosecurity highlighted that the combination of **order and customer screening** to arrive at a decision is more likely to require machine learning techniques and AI because it is too complex for a human actor to be able to detect sequences of concern, especially if there are any attempts to obfuscate the sequences, e.g. slight twitches to nucleotides that would obfuscate the order. Indeed, several experts coming from academia, government and industry highlighted the potential for AI to go **beyond list-matching** when looking at sequences given that list-based approaches may have limitations when considering new threats. According to several experts from academia and policy, new AI designs can make proteins that have the same structure and therefore have the same function but have a very different primary DNA sequence. AI in synthetic nucleic acid orders screening can be useful to detect genetically engineered organisms that are not present in known lists or whether there is any kind of potential harm in a particular sequence (better understanding of the function, identification of obfuscation, etc.).

A DNA synthesis provider also highlighted **how the screening of orders works for a company in practice**: customers order to the company, they send the sequence over secure encrypted channels to the screening provider, which generates threat assessment data and sends it back to the data synthesis company that reviews the information to make final decisions. The **automated system cannot make a final decision**, it only generates data to support human decision making. At the end of the day, a **human must decide** whether something crosses a threshold in terms of having to follow up with the customer, asking them what they're doing with the order or if it requires an export license.

Policy and academic experts shed further light into the functioning and development of the **common mechanism under the International Biosecurity and Biosafety Initiative for Science (IBBIS)**. The project came out of Nuclear Threat Initiative, a nonprofit global security organization focused on reducing nuclear and biological threats imperilling humanity, which set up an international working group with people coming from the policy side, from the gene synthesis industry and from the bio security community. The mechanism is not doing a direct sequence comparison, but it uses a probability matrix for any given amino acid being encoded within that genetic information. Instead of looking at the sequence of concern, it has developed Hidden Markov Models**Error! Bookmark not defined.** of specific sequences that tell you which amino acid residues are important for maintaining the structure and the function of that sequence. The common mechanism restricted its database to a smaller set of sequences that they focus on, i.e. only things that are found in regulated pathogens on already published and well-established lists of sequences of concern.

Challenges

Both academic, government and policy expert working on these topics underlined that it is inherently **challenging to integrate AI and biology** projects because there are very few people who are truly experts in both fields. Several of the interviewed experts working on these topics struggled to find European experts in the field who deal with AI, biotechnologies and security.

With regards to the **challenges linked to screening**, several experts working in the industry agreed that the key challenge is not technically doing the screening, but **agreeing on what order should be fulfilled and the criteria** for fulfilling that order. Currently there is no certification process yet and several experts reported that NIST is working on establishing it in the US. Few policy experts highlighted that a **barrier to the use of screening tools** is the fact that companies might be worried that it makes them liable for misuse. Several providers reported that it is extremely rare for them to reject an order, and this has become part of the problem with biosecurity screening. One industry expert mentioned that, within the IGSC, they have a mechanism for **order reporting**, but it's been used once since 2009, and the challenge resides in the fact that companies are reticent about disclosing customers due to competition. Internal discussions have started with respect to the possibility of having a neutral third party to report to or some blinding method where one submits that information to a mathematical process and then if another party submits the same information, then it's unblinded.

One of the **challenges of AI for synthetic nucleic acid**s screening detection mentioned by both industry and academic experts is that a malicious actor could **order nucleic acids in pieces** and defeat all screening. This means that there is a trivial way by which any educated adversary could get around it and good enough de novo protein design will eventually obviate synthetic nucleic acid screening. One industry representative highlighted that synthesis companies are worried about integrating AI into their models as they may run into the same **unpredictability seen in AI models within other domains,** i.e. do not want to risk that 5% of the time the model may miss or forget some part of the embedding space. At the same time, experts mentioned that companies don't want all the developments in this field to be fully known because they create **vulnerabilities for companies**.

An academic expert mentioned that a greater focus amongst tool providers should be to make **screening robust to AI circumvention as opposed to applying AI directly to build better tools**. Afterwards, there will be a longer-term aim to figure out how to use AI tools inside the screening tools themselves to develop more sophisticated approaches to screening and have a better set of data against which to screen. The implementation of a good tool would be **expensive** as it would require **in-depth testing**.

With respect to benchtop devices, one government expert underlined that one option would be to have **machines equipped with software and algorithms** preventing them from synthesizing sequences that can have a dangerous function.

New Regulations in place

Academic experts and industry representatives also provided their view on the United States executive actions to require that federally funded projects are from a company that screens orders. It was highlighted that different groups are covered by this, i.e. not only DNA and RNA providers but also contract research organizations working for pharma companies, university core facilities, or synthesizer companies that make desktop machines. According to both academia and industry, this regulation is a **very low bar**, but it's a first step towards the general acceptance of standards within the industry. Indeed, they underlined that it is a **voluntary statement** that the company must make saying that they meet the criteria. According to several experts, **AI has been a driving threat to allow people to become aware** of the fact that it would not be possible to defend against

the misuse of AI unless these other things are implemented. Academic and industry experts are expecting the biggest uptake of screening tools over the next couple of months with the change in the US rules. They also highlighted that two years from now there will be a **certification process** in place based on the activities by the IGSC Working group. Indeed, through IBBIS there has been broad collaboration across screening tool providers on benchmarking, to better understand the individual strengths and weaknesses of the different tools.

2.4. AI-enabled tools to improve the detection and attribution of genetic engineering

2.4.1. AI-based tools to enhance the detection of genetic engineering

These are the main takeaways on the role of AI-based tools to enhance the detection of genetic engineering.

- The study identified three tools, two of which are AI-based, currently being used or in development for detecting genetic engineering (Plaster, Synsor and ENDAR).
- Interviews with experts were conducted to evaluate the current AI tools used for genetic engineering detection.
- The identified tools employ deep learning techniques (Synsor) or alignment-free methods (ENDAR).

Our study highlights the need for integrating high-quality DNA sequence datasets to train deep learning models and suggests a combined approach using multiple detection methods to enhance detection rates.

Current developments in synthetic biology facilitate the generation of novel biological components. These novel biological components pose a risk to public health either by an accidental introduction into the environment or the possible misuse or abuse by malicious actors. Therefore, to safeguard security and public health, **it is essential to monitor the presence of genetic engineering in biological systems.** This is a promising approach for an early detection of emerging biothreats. Different computational and experimental tools and protocols have been developed to identify signals of genetic engineering. However, the characterisation of genetic engineering remains challenging (Tay, et al., 2024).

New methods are needed to enable a more targeted approach to identify engineered DNA in biological systems. This study has identified three tools currently being used or under development for detecting genetic engineering. The tools are presented in detail below and summarised and compared in Table 4.

Wang and colleagues (2019) **developed an efficient and explainable method to detect signatures of genetic engineering in plasmids**. Plasmids are small, circular, DNA molecules that are easy to modify and commonly used in biological research to transfer novel functions between species. Their method is based on the “pan-genome” concept, where a pan-genome sequence is a compact reference sequence capturing all the existing variation in a group. To achieve this, they developed a new pan-genome construction algorithm called Plaster. The Plaster algorithm generated two reference pan-genomes: the synthetic one capturing the diversity from 51,047 synthetic plasmid sequences, and the natural one capturing the diversity from 6,642 natural plasmid sequences. By comparing unknown plasmid sequences to these reference pan-genomes, the method can distinguish whether a plasmid is natural or engineered. Their research demonstrated that the Plaster algorithm generated pan-genome sequences that permitted the differentiation of synthetic and natural plasmid sequences (Wang, Elworth, Liu, & Treangen, 2019). **This approach did not integrate any ML**

algorithms, reinforcing the explainability of the model. The authors described that their initial approach to genetic engineering detection would lead to new techniques and approaches for rapid characterization of engineered plasmids.

Tay and colleagues (2024) **developed a tool for alignment-free detection of engineered DNA sequences using an artificial neural network.** The tool relies on the differences in oligonucleotide frequency (k-mer signature) between naturally occurring and engineered DNA sequences. Tay and colleagues developed Synsor based on the assumption by Allen et al. (2008) that engineered DNA sequences contain k-mer signatures notably different to natural occurring sequences**. Synsor uses artificial neural networks that capture complex non-linear relationships and can classify whether a query DNA sequence is natural or engineered through pattern recognition of k-mer signatures.** Synsor is one of the first tools designed for genetic engineering detection whose capacity to detect potential genetically engineered organisms has been demonstrated with over 99% accuracy and which does not rely on sequence alignment, opening the way for new methods using deep learning and artificial neural networks (Tay, et al., 2024).

In addition to academic tools, the literature highlights IARPA's FELIX Project (Finding Engineering-Linked Indicators project), which aims to enhance biodetection and biosurveillance by developing new methods to identify indicators of genetic engineering (Lewis, et al., 2020). One solution funded under this initiative is **ENDAR** (Engineered Nucleotide Detection and Ranking), a software tool developed by Ginkgo Bioworks, a Boston biotech company, and Draper, a nonprofit engineering innovation firm. **ENDAR uses deep learning to help trained analysts identify genetic engineering in next-generation sequencing (NGS) data**. It is designed to work across various biological organisms found in complex, multi-species environments. According to its developers, ENDAR significantly improves the signal-to-noise ratio compared to previous methods, which could allow for earlier detection of engineered organisms. The project's outcomes were announced in a press release in 2022 (Kelly, 2022), but no scientific publication has yet detailed their findings and technical aspects of the developed machine learning solution.

Table *4* AI-based tools identi¸ ed for the detection of genetic *engineering.*

Name	AI technique	Type of genetic material analysed	Accuracy	Target	Availability	Source
Plaster: pan-genome construction algorithm	Alternative to deep learning - Basic clustering methods	Addgene database (plasmids)	Information not available	Detection of GE	For research [source code publicly available]	Wang et al., 2019
Synsor	Deep learning - artificial neural network	Addgene database (plasmids)	>99% accuracy	Detection of GE	For research [source code available under GPL v3 license]	Tay et al., 2024
ENDAR	Deep learning	Information not available	Information not available	Detection of GE	On the market by Ginko bioworks	Unpublished

Source: *author's elaboration based on literature review findings.*

The literature also considers the **key challenges and considerations for using AI models to detect genetic engineering**. It emphasises the need for high-quality DNA sequences to train deep learning models and the trade-off between model performance and required computational resources. Some authors propose combining multiple detection methods to improve detection rates (Tay, et al., 2024) (Crook, et al., 2022). Additionally, concerns are raised about open-source tools potentially aiding individuals in designing genetic elements that

could evade detection. Some authors advocate for developing more comprehensive, closed-source versions of the tools (Tay, et al., 2024).

2.4.2. AI-based tools to enhance the attribution of genetic engineering

These are the main takeaways on the role of AI-based tools to enhance the attribution of genetic engineering.

- Identification of six AI-based tools for genetic engineering attribution. The most used AI technique is Convolutional Neural Networks, with employing traditional clustering methods and Random Forest.
- Interviews with experts in the field validated the identified challenges such as uncertainty about real-world performance and the tools' lack of explainability, that might hinder adoption and action upon their results.
- Experts interviewed agreed on the importance of considering these methods as only one piece of information that should be combined with other evidence or surveillance.
- All the identified tools were mainly developed between 2018 and 2022, but progress has stalled. Experts suggest that the commercial potential of the tools (i.e., as a solution for intellectual property rights in genetic engineering) could act as a catalyst for future developments.

The previous section describes the different tools and methods developed for the detection of genetic engineering. Other approaches aim to improve biosecurity by the attribution of genetic engineering to the source laboratory or designer. **The ability to determine who is the responsible actor of a human-caused biological could be detrimental to the misuse of genetic engineering** (Alley, 2021) (Lewis, et al., 2020) (Carter et al., 2023a).

The attribution of genetic engineering relies on the idea that scientists who perform genetic modifications leave behind unique 'methodological signatures'. These signatures come from specific design choices, such as a preference for certain genetic elements or specific codon optimization methods, along with evolutionary markers. These signatures can be used to infer attribution of the genetic construct (Lewis, et al., 2020). Because the data is vast and complex, advanced analysis is necessary. **To meet this need, researchers have developed machine learning techniques** (Alley, 2021). The tools are described below and summarised and compared in Table 5.

In late 2018, Nielsen and Voigt (2018) demonstrated that deep learning could be used to interpret engineering methodological signatures to predict the laboratory of origin of genetic engineered DNA sequences. In this first attempt, a **convolutional neural network (CNN)** was trained on the Addgene dataset that at the time contained engineered plasmids from 2230 laboratories. The trained CNN was used to establish the origin of DNA sequences. The accuracy was of the 48% for of the laboratory-of-origin prediction but increased up to a 70% for the top 10 predicted laboratories. This method, on average, allowed for variations in ~9% of the sequence without a substantial reduction in the predictive accuracy (Lewis, et al., 2020) (Nielsen & Voigt, 2018). This work showed that **genetic engineering patterns can guide attribution without manual feature selection or sequence function data**, opening the door for future research.

Two years later, Alley and colleagues (Alley, 2021) proposed the use of **recurrent neural networks (RNNs)** over CNNs for their suitability in detecting and predicting patterns in sequential data and handling variable-length sequences. The team used techniques from natural language processing (NLP) to extract features from the sequence. More specifically, Byte-Pair Encoding (BPE) algorithm was used to infer 1000 salient motifs directly from the Addgene primary DNA sequences originating from over 1400 labs. These motifs were used for

the training of the RNN together with some basic phenotypic information which improved the accuracy of the model. The resulting model**, deteRNNt,** reached 70.1% accuracy for the prediction of the laboratory of origin and 84.7% top 10 accuracy (Alley, 2021). To enhance usability, the team developed a framework to weight predictions against other investigative evidence improving robustness and uncertainty estimation for integration with other attribution indicators. They also demonstrated that simple classification models can accurately predict the nation and region of origin and "genealogies" of laboratories. This is particularly relevant for the Biological Weapons Convention (BWC) and to expand the identification of designer laboratories beyond those in the training dataset (Alley, 2021). These methods resulted in the development of an **integrated attribution toolkit of machine-learning algorithms, designed to enhance biosecurity and support biotechnology stakeholders in promoting responsible innovation** (Alley, 2021) (Lewis, et al., 2020). Despite its improved accuracy, the toolkit relies on high-quality training data and while Addgene attribution data provide a reasonable model scenario, it is biased because of the limitation of plasmid sequences and its geographical location in the US. Additionally, machine learning models can be costly and lack explainability, often obscuring the features and decision-making process generating a given classification or prediction (Holzinger, 2018).

In parallel to the development of the deteRNNt model, Wang and colleagues (2021) developed **PlasmidHawk, an alternative alignment-based approach for laboratory-of-origin detection independent of machine learning techniques**, overcoming some of the limitations presented above. PlasmidHawk builds a pan-genome from the synthetic plasmid training data (Addgene) using Plaster, the pan-genome construction algorithm presented above (Wang, Elworth, Liu, & Treangen, 2019). The generated synthetic pan-genome is annotated, connecting unique regions with information on laboratory of origin. PlasmidHawk can predict the laboratory-of-origin of unknown plasmids by performing sequence aligning between the query unknown sequences and the annotated synthetic pan-genome. This approach led to an accuracy of 76% for top 1 prediction and 85% accuracy in the top 10 predictions (Wang, Kille, Liu, Elworth, & Treangen, 2021). This study demonstrated that alignment of query sequences to specifically designed pan-genomes can effectively predict laboratory-of-origin of synthetic sequences while providing interpretable prediction results, a benefit compared to the currently limited explainability of ML-based methods. **The authors highlighted the importance of continuing the study of traditional comparative genomics methods to complement the more recent advances in AI and ML methodologies**.

Based on the alternative approach to the laboratory-of-origin issue proposed by Wang et al. (2021), researchers that worked on the development of the deteRNNt collaborated in the organisation of **the first Genetic Engineering Attribution Challenge (GEAC) to explore additional and improved solutions to the matter** (Crook, et al., 2022). The citizen science initiative resulted in 40 teams outperforming the top 1 prediction accuracy, while over 70 teams improved the top 10 prediction accuracy of laboratory-of-origin and compared to the existing models at the time. Some of the most promising results included a nearly 95% accuracy for top 10 prediction and 81.9% for the top 1 prediction using a **CNN-based approach**, and a 1000-fold increase in speed by a high accuracy method based on a **naive Bayes classifier** instead of CNN (Crook, et al., 2022). Despite the encouraging promising results of the competition, the GEAC did not focus on the calibration of the models, and therefore, the resulting models were much less well-calibrated than the deteRNNt model developed by Alley et al. (2021). The study highlights the gap between the competition-based attribution challenge and real-world GEA applications. While the Addgene database offers a good testing ground, its simplicity limits real-world applicability, requiring further research to generalise attribution models (Crook, et al., 2022).

In parallel, in 2022, a different research group introduced a new approach to GEA. They developed a state-of-the-art classifier and a training methodology for interpretable GEA. Their method uses **deep metric learning**, which teaches the model to learn the similarity between

plasmid sequences and the information from the laboratories. They also improved the use of **CNNs by employing circular shift augmentation.** This technique helps the model extract DNA embeddings while learning the embeddings of known laboratories from the Addgene database. This training methodology improves the accuracy of the model. The embeddings can be used as features in other ML models and perform downstream tasks like clustering DNA sequences and laboratories. The authors also introduce the use of triplet networks to embed the DNA sequences from laboratories and DNA query sequence. The use of triplet networks allows multiple comparisons beyond line ranking the possible laboratories-of-origin, opening the door for multiple clustering techniques and improving interpretability of the model. Increased robustness to sequence perturbations was achieved compared to previous attempts (Nielsen & Voigt, 2018), allowing as far as nearly 1000 variations in the sequence while maintaining an acceptable accuracy in the top 10 prediction. Moreover, this method offers an efficient approach that avoids retraining the entire model when adding new laboratories. Instead, it directly analyses the distance between an unknown sequence and the new laboratories embeddings (Soares, Camargo, Marques, & Crook, 2022).

Spirgel and colleagues (Spirgel, Comolli, & Guido, 2023) developed a **methodology that addressed attribution from the signatures left by the computational tools used to design the engineered modifications**. The team developed a **Random Forest (RF) classifier** that distinguishes between the design tools used for codon optimisation. This methodology achieved a 97% accuracy in predicting the computational tools used, a piece of information useful for the attribution of GEA and groundwork for future research lines, as stated by the authors of the study.

Table *5* AI-based tools identi¸ ed for the attribution of genetic engineering.

Name	AI technique	Type of genetic material analysed	Accuracy	Target	Availability	Source
Deep learning to predict the laboratory-of-origin of engineered DNA	Convolutional Neural Network (CNN)	Addgene (plasmids)	47% top1 70% top10	Laboratory-of-origin	For research	Nielsen and Voigt, 2018
ML toolkit including deteRNNt model for laboratory-of-origin attribution	BPE for plasmids encode; Recurrent Neural Networks (RNNs)	Addgene (plasmids)	70% top1 85% top10	Laboratory-of-origin; Nation-of-origin; Ancestor laboratory-of-origin	For research	Alley et al., 2021
PlasmidHawk (integrating Plaster algorithm)	Alternative to deep learning - Basic clustering methods	Addgene (plasmids)	76% top 1 85% top 10	Laboratory-of-origin	For research	Wang et al., 2021
Results from the GEA challenge	CNN and naïve Bayes classifier	Addgene (plasmids)	81.9% top 1 94.9% top 10	Laboratory-of-origin	For research	Crook et al., 2022
Deep metric learning to improve laboratory-of-origin prediction	Deep metric learning and CNN with circular shift augmentation	Addgene (plasmids)	75.8% top 1 90.4% top 10	Laboratory-of-origin	For research	Soares et al., 2022
ML for Genome Engineering Design Tool Attribution	Random Forest classifier	Information not available	97% top 1	In-silico tools used to design codon optimization	For research	Spirgel et al., 2023

Source: *author's elaboration based on literature review findings.*

Despite advances in the field, **there are some considerations and limitations that need to be considered when analysing these AI-based techniques to enhance GEA**. First, attribution techniques are not intended for the detection of genetic engineering and therefore GEA is a process that should follow the initial process of detecting engineering (Lewis, et al., 2020). Second, these approaches do not provide authenticity guarantees, as laboratories could vary their preferred methods or third parties could mimic methodological signatures, difficulting the origin identification. While useful for initial investigations, genetic analysis through Computational and AI-based tools should be seen as a supplement to GEA, always accompanied by non-technical indicators like context, intelligence from human sources or surveillance data (Nielsen & Voigt, 2018) (Lewis, et al., 2020) (Alley, 2021) . Third, most models are trained using Addgene which, despite its large representation of synthetic genetic sequences, does not represent the real world, especially when considering GEA for intended misuse where actors will look for ways of challenging attribution indicators (Lewis, et al., 2020) (Alley, 2021).

2.4.3. Considerations from the experts

The findings in previous sections were complemented through the consultation activities with key experts. Experts' perspectives have contributed to a better overview of the state-of-the-art on genetic engineering detection and attribution. Box 5 presents the main learnings from the interviews conducted with several experts belonging to Academia, Industry, Government, and International Agencies.

Box 5 *Considerations from the experts: AI-enabled tools to improve the detection and attribution of genetic engineering*

Highlights resulting from Experts' consultation

Several industry, academic and policy experts working in this field were consulted to gather which are the main opportunities and challenges for incorporating AI into genetic engineering detection (GED) and attribution (GEA), and therefore, enhance the biosecurity framework.

Challenges in the use of AI for enhancing GED and GEA

One major challenge in the use of AI for GED and GEA lies in the **explainability of AI systems, particularly when attempting to prompt actionable responses from authorities and governments**. An academic expert emphasized that the necessary level of confidence in AI-driven findings varies depending on the intended application. For instance, if the goal is ongoing surveillance of an individual or entity, moderate confidence may be sufficient, whereas attributing a significant action to a nation-state may require a much higher confidence level before making a public statement.

Another complication arises from the gap between AI models and their real-world use cases; as several academic experts noted, tools often depend on the presence of specific data in training sets, making it difficult to anticipate performance in practical applications without real-world testing. Several experts noted that AI-based tools for GED or GEA are strengthened when they are integrated with other types of evidence, such as location data, video footage, or physical traces from a scene, aligning with realistic investigative needs.

An academic and an international organization expert highlighted that existing GEA methodologies, largely based on the Addgene dataset, lack relevance to real-world applications like identifying bioweapons. Current attribution datasets involve plasmids that bear little resemblance to actual bioweapons, and those synthesizing the plasmids in these test sets are not typically seeking to evade detection. As an academic noted, these factors create a gap between GEA's current capabilities and its potential to identify those responsible for creating bioweapons. Should malicious actors aim to evade attribution, they might mimic other engineering styles or capitalize on discrepancies

between training data and bioweapon characteristics, complicating attribution efforts considerably. Thus**, challenges persist due to the unrealistic nature of current datasets and the evolving tactics that could be employed by adversaries**.

An academic expert working in industry emphasized that **GEA is a rapidly evolving field which poses a formidable challenge**; the advancements seen today will likely differ significantly within five to ten years as more tool-assisted engineering practices replace manual laboratory work. Attribution is an area where fundamental research from academia is essential, yet current research often serves only as a preliminary step toward actionable solutions.

Industry perspectives also emphasize that the **absence of commercial incentives and limited public funding in biosecurity** means that many AI and machine learning professionals are drawn to more profitable sectors.

Opportunities in the use of AI for enhancing GED and GEA

Academic and policy experts noted that **GED and GEA tools have a strong deterrent effect.** The possibility that AI-based detection and attribution tools could expose malicious actors, trace their actions, or help mitigate impacts swiftly may be sufficient to deter harmful intent.

Some experts reflected in a couple of key outcomes from the GEA Challenge, proving that **natural language processing (NLP) techniques can be valuable for genomic analysis** and that AI experts with well framed problems and questions may have a crucial role in developing solutions to enhance biosecurity, despite their lack of knowledge in the biosecurity field.

Another expert recalled some initiatives on developing **guardrails for biodesign tools that incorporate metadata tags, potentially cryptographically signed, to document design origins, user intentions, and optimization steps**. This could allow DNA providers and collaborators to trace biodesigns and interpret their intended functions, enhancing both security and transparency. Another expert further proposed the **"digital barcoding" of synthetic DNA, embedding unique identifiers in sequences to trace their origin**, which could aid forensic investigations and compliance. This concept may not require intensive AI-specific technology but **could be accelerated by AI to handle larger volumes of data**.

Latest developments and next steps

Industry experts further noted that **GED and GEA serve as supplementary intelligence sources within a broader biosurveillance ecosystem**, akin to the chemical weapons surveillance landscape. Expanded biosurveillance programs, such as wastewater monitoring, offer granular environmental data that could support GED and GEA's practical application, yet **scaling such initiatives would require substantial investment**. Despite their potential, such programs remain limited, hindering the broader integration of GED and GEA as mainstream biosurveillance tools.

Several experts also noted that in the near future, **GEA might raise some emerging commercial interest**, which would accelerate its development, as it could potentially offer solutions to Intellectual Property concerns and the potential risks related to intellectual property theft and patent violations.

From an industry perspective, **having diverse, independent solutions is essential to GED and GEA long-term success**, as consensus-driven approaches from unbiased sources add credibility and enhance international acceptance.

Consequently, though GED and GEA are a promising area for AI application, some technical, practical, and economic obstacles complicate its advancement.

2.5. Outlook and potential future applications of AI

These are the main takeaways on potential future applications of AI in biosecurity.

- Beyond its use for synthetic nucleic acid orders screening, GED and GEA, AI has potential in other areas to safeguard biosecurity such as biosurveillance and early warning systems, threat detection, response strategies, and establishing mitigation measures.
- Large Language Models and Generative AI are considered by many as underexplored tools that could help advance biosecurity.

AI can enhance the screening of synthetic nucleic acid orders and can expedite the detection and attribution of genetic engineering. While these advances are extremely relevant, **the literature review revealed other areas that need to be considered when discussing the role of AI in biosecurity.**

Drawing on insights from the study of De Haro (2024) on the use of Artificial Intelligence in synthetic biology, the outcomes of the meeting "Exploring possible impact of AI on Biosecurity and International Cooperation in the BWC" (InterAcademy Partnership, 2024) and the report on "the convergence of AI and the life sciences" (Carter et al., 2023a), **Table 6** Error! Reference source not found. **provides and overview of AI-driven innovations applied to biosecurity**.

The table presents the potential contribution of AI systems and tools across three main steps in biosecurity: (1) AI for biosurveillance, data integration, and early warning systems; (2) AI for biothreat detection and monitoring; (3) AI for biothreat response and mitigation. Practical examples of existing tools or use cases are highlighted when available.

This overview of the potential application of AI innovations in biosecurity, showing how AI supports efforts to safeguard public health and mitigate biological risks.

Table 6 Opportunities of AI to synthetic biology and biosecurity

AI for biosurveillance, data integration and early warning systems	
Area of application	**Description of AI contribution**
Data analysis	AI has been a significant change in the analysis of genomic data, accelerating the pace of discovery in synthetic biology. The volume of data produced by modern genomic technologies such as Next-Generation Sequencing is immense. AI algorithms, particularly machine learning and deep learning models, can learn from and make predictions on these rapidly, making the analysis possible on the scale of hours todays. **Tool example**: ART (Automated Recommendation Tool). ART leverages machine learning and probabilistic modelling techniques to systematically guide synthetic biology without the need for a full mechanistic understanding of the biological system. (Radivojević, Costello, Workman, & Martin, 2020)
Biorisk management	AI can predict the safety and security implications of certain research activities. For instance, it could help predict the likelihood of an engineered organism's accidental release or identify research activities that could be misused.
Monitoring of biotechnology activities	AI can be used to monitor biotechnology activities and to verify and validate information, ensuring safety and beneficial outcomes from biotechnology and pharmaceutical research and development. AI-based tools can facilitate the identification of patterns of unapproved acquisitions, new life-science discoveries, and interesting and potentially concerning publications for further review.
Biosurveillance and detection	AI can support early detection of biological threats and outbreaks combined with environmental or human surveillance systems by the analysis of patters and deviations from normality. An example of this is the use of AI to enhance metagenomic analysis for genomic

AI for biosurveillance, data integration and early warning systems	
Area of application	**Description of AI contribution**
	detection in wastewater surveillance. AI can also be used to analyse the raw data, flagging concerning inputs for human review.
Modelling patterns of infectious diseases and Outbreak response	AI tools can be used for monitoring and tracking patterns of infectious disease outbreaks, predicting and preventing the spread of disease, and processing and using the information to reduce disease impact, evaluate risks to public health, and inform policy decisions and containment measures. AI tools can test response capabilities, robustness of oversight mechanisms, and safety of biotechnology products and services. Together with other new technologies, AI can contribute to control and securing supply chains which is of significant relevance for pandemic preparedness and response.
Reducing misinformation and disinformation	AI tools might help in the detection and recognition of misinformation and disinformation, as well as in the summarisation and outreach of confidence building measures created at governmental level.

AI for biothreat detection and monitoring	
Area of application	**Description of AI contribution**
Synthetic nucleic acid screening	AI can be used for nucleic acid synthesis screening to help prevent the unauthorized creation of harmful organisms. **Tool example**: Tools reviewed in Section 2.1 of this report.
GED and GEA	AI can enhance the process of genetic engineering detection and attribution. **Tool example**: Tools reviewed in Sections 2.4.1 and 2.4.2 of this report.
Protein structure prediction	Protein structure prediction is one of the key applications of AI in synthetic biology. Understanding how a protein's function is determined by its 3Dstructure is one of the central challenges in biology. This is because the number of configurations a protein can fold into is astronomically large, making it practically impossible to predict a protein's structure based solely on its sequence of amino acids using conventional methods. AI can help accurately predict protein folding based on sequences **Tool example**: BDTs like RoseTTAFold, AlphaFold-2, ProteinMPNN, ESMFold (Carter et al., 2023a)
Library screening	AI can help screen libraries of various biological or chemical entities, significantly improving the speed, cost-effectiveness, and outcomes of library screening.
Forensic investigations	AI and especially ML can contribute to genetic engineering detection and attribution in forensic investigations, as described in the previous sections of the report. AI tools can monitor attempts to create a new pathogen, or to build a known high-risk pathogen, allowing controls to be put in place for biotechnology oversight.

AI for biothreat response and mitigation	
Area of application	**Description of AI contribution**
Vaccine and medicine development	AI has been pivotal in accelerating various stages of vaccine development, including antigen identification, vaccine design, production, and distribution. Moreover, AI can also speed up the process of medicine screening by analysing large databases of compounds and predict their likely effects, reducing the need for extensive laboratory testing

AI for biothreat response and mitigation	
Area of application	**Description of AI contribution**
	Tool example: SAM (Search Algorithm for Ligands). SAM is an AI program that shifts through trillions of molecules to determine which one could be a good vaccine adjuvant candidate (Masige, 2019) (Lima, Cha, Cha, & Hwang, 2021).
Gene editing	AI can be used to predict the outcomes of genetic modifications using specific tools (e.g. CRISPR). By analysing large datasets, AI can predict off-target effects and help optimize the design of guide RNAs, leading to more efficient and accurate gene editing. **Tool example**: Combination of AI and CRISPR/Cas9 (Maserat, 2022)
***De novo* gene design**	AI can be used in designing and creating new genes from scratch to perform a specific function in various applications, e.g. medical treatments. **Tool example**: A generative model that bridges systems biology and molecular design, conditioning a generative adversarial network with transcriptomic data (Méndez-Lucio, Baillif, Clevert, Rouquié, & Wichard, 2020)
Genetic circuit design	Designing genetic circuits, which are sequences of DNA that enable cells to perform new functions, is a complex task. AI can help to design these circuits and predict how they will behave in living cells **Tool example**: machine learning and neural networks for understanding the relationship between motif choice, combination and position to tailor synthetic receptors for desired cell phenotypes (Daniels, et al., 2022).
Protein design	AI can help to predict how changes in a protein's sequence will affect its structure and function. This can speed up designing new proteins or modifying existing ones **Tool example**: RFDiffusion
Automated laboratory experiments	AI can guide the automation of laboratory experiments. This can greatly speed up the research process and make it more efficient. **Tool example**: Emerald Cloud Lab, Opentrons
Risk assessment on biothreats	AI can enhance a more robust and informative risk assessment on biothreats, aiding in real-time those in charge of decision-making.
Supply chain management (Outbreak response)	AI controls supply chains relevant to pandemic preparedness and response.
Timely diagnosis and patient management	The detection of specific symptomatology, timely diagnosis of infection, assistance in patient triage, assessment of the illness prognosis and patient's response to treatment, and the monitoring of population's adherence to public health recommendations.

Source: *Adapted from De Haro* (2024)*, the meeting Exploring possible impact of AI on Biosecurity and International Cooperation in the BWC* (2024) *and the report on the convergence of AI and the life sciences* (2023a).

3. Discussion & Conclusion

AI offers significant potential to strengthen the EU biosecurity framework, ranging from improved biosurveillance and early warning systems to enhancing biothreat response and mitigation. However, these benefits require coordinated action at multiple levels. Building on existing EU and international biosecurity measures, a series of targeted actions can help translate AI's capabilities into concrete safeguards. The following section outlines **key measures the EU can adopt to harness AI for enhanced biosecurity while ensuring ethical use, regulatory alignment, and cross-border cooperation.** This chapter concludes by summarising the key conclusions and insights of the study and providing a comprehensive **overview of their alignment with the original objectives**.

3.1. Recommendations to leverage AI for an enhanced EU biosecurity framework

The uncertainty surrounding the new developments in both AI and synthetic biology demands that all stakeholders take a proactive approach to mitigate risks and leverage the opportunities these two emerging fields present for public health (Carter et al., 2023a). Experts interviewed in this study emphasized the urgency of the moment. As the intersection of AI, synthetic biology, and biosecurity is relatively new, there is a unique opportunity to shape its future by integrating safety by design into policies and tools. However, this window of opportunity is closing rapidly, prompting calls for swift international action.

The following globally scoped recommendations can serve as a practical roadmap to guide the EU's direction in implementing AI's potential to enhance the EU biosecurity framework, as outlined in the previous section.

Promote multidisciplinary and cross-border collaboration in the fields of AI and biosecurity

To maximise the application of AI in biosecurity, experts emphasise the need for a multidisciplinary approach involving microbiology, synthetic biology, computational sciences, public health, and bioethics professionals (Carter et al., 2023a) (InterAcademy Partnership, 2024) (Undheim, 2024). Sustainable growth of computational technologies applied to life sciences like Biodesign tools and LLM requires collaboration among multidisciplinary teams to ensure a good alignment between the technical, ethical, and security implications of these advancements (Sandbrink, 2023). The EU can foster dialogue between industry, academia and policymakers by supporting scientific associations. Such multidisciplinary groups are essential to ensure diverse perspectives when discussing the implications of new technologies' development (Urbina, Lentzos, Invernizzi, & Ekins, 2022). Proposed measures include:

- **Participate in Global AI-Biosecurity Initiatives**. Take a leading role in international discussions on AI and biosecurity, ensuring that EU actions align with and influence global governance frameworks.
- **Reinforce EU multidisciplinary bodies to advise governments on AI applied to the life sciences and biosecurity**. Proposed initiatives include a BWC scientific advisory body to engage with the latest developments and identify potential areas of concern or interest for the BWC (InterAcademy Partnership, 2024) and an international "AI-Bio Forum" for sharing best practices and recent developments in AI and biosecurity.
- **Promote EU programmes and discussion forums to unite key experts to shape policies and guidelines for an enhanced and secure interplay between AI, synthetic biology, and biosecurity**. The European Biosecurity Regulators Forum (EBRF), comprising Belgium, Denmark, France, Germany, The Netherlands, Sweden, Switzerland, and the UK, serves as a solid foundation for **establishing a European network of biosecurity contact points**, with could expand to other Member States with proper EU support.

Support Research, Innovation and Capacity building in AI for Biosecurity

Both the literature reviewed, and the experts interviewed highlighted the importance of the allocation of resources for research and innovation applied to AI in the context of enhancing the biosecurity framework. Specific measures proposed include:

- **Strengthen EU's research agenda by funding AI tools for pandemic preparedness and response through programmes like Horizon Europe and EU4Health**. Suggested focus areas for these programmes include enhanced screening

of synthetic nucleic acid orders, early outbreak detection, accelerated vaccine development, and improved AI control mechanisms to safeguard security.

- **Increase government funding to develop robust biosecurity infrastructure**, ensuring the capacity to identify the origin, danger level, and spread of biological threats, while addressing the lack of private sector incentives due to limited profitability.
- **Support the creation of data-sharing protocols for AI model development and testing.** Controlled access to the data required for training, auditing and evaluating AI models would enhance responsible use and minimisation of risks while enabling innovation and continuous progress (Sandbrink, 2023) (Urbina, Lentzos, Invernizzi, & Ekins, 2022).
- **Enforce an assessment by scientific publishers of the data and information made publicly available in scientific publications** for adequacy and potential risks assessment (National Security Commission on Emerging Biotechnology (NSCEB), 2024).

Moreover, it was highlighted the importance of enhancing understanding of biosecurity implications among life sciences and computer science professionals (Urbina, Lentzos, Invernizzi, & Ekins, 2022). **Interdisciplinary training that integrates ethical, social, computing, and life sciences can improve awareness of both the benefits and potential misuse of new advancements, fostering greater understanding and trust**. While education policies are mainly handled by Member States, the EU could support and facilitate access to biosecurity education initiatives by developing a code of conduct for biosecurity similar to the ones used for biosafety and promoting the common EU education frameworks and mobility programmes that expand cross-border training opportunities for students across Member States.

Develope Standards and Guidelines for the use of AI and biotechnology in Life Sciences

In order to prevent misuse, it is essential to establish robust standards and guidelines for ethical and practical challenges posed by emerging technologies. The literature and interviewed experts proposed several roles that the EU could take.

- Collaborate with Member States to **adopt a biotechnology and AI code of conduct, similar to The Hague Ethical Guidelines for chemistry** (Organisation for the Prohibition of Chemical Weapons (OPCW), 2015), to promote responsible use and reduce risks of misuse (Urbina, Lentzos, Invernizzi, & Ekins, 2022).
- **Establish international agreements on regulated pathogen lists and develop guidelines and protocols for AI-enhanced screening synthesis orders tools for other biological threats,** such peptide toxins, that can be synthesized in low-biosecurity environments (Lee, Cowan, & & Tankard, 2022).
- **Promote, in collaboration with other international regulators, the development and implementation of standards specific to the use of AI models in life sciences**. The development of standards for the use AI models is a global matter, extending beyond their application in life sciences. Initiatives like the European AI Act, OECD AI principles, and ISO standards aim to enhance transparency, data quality and system reliability and are the starting point of any further initiatives.

  Establish oversight and internationally recognized standards to tackle visibility and traceability issues arising from decentralised nucleic acid synthesis using benchtop devices (Carter & Yassif, 2023b) (Kane & Parker, 2024). Existing frameworks, such as ISO 35001:2019 on “biorisk management for laboratories” and ISO 20688-2:2024 on “Biotechnology - Nucleic acid synthesis - requirements for the

production and quality control of synthesized gene fragments, genes, and genomes," serve as a basis for developing targeted standards in the evolving biotechnology sector.

Establish rules and legislation for AI in the life sciences to improve EU's Biosecurity

Legislative frameworks are being put in place worldwide for the secure development and use of AI technologies. Two examples are the Artificial Intelligence (AI) Act in the EU and previous US Executive Orders on AI, including EO 14110 on Safe, Secure and Trustworthy Development and Use of Artificial Intelligence. While the 2023 US Executive Order on AI marked an initial move towards enhanced biosecurity, these regulations are generally unspecific concerning the application of AI to the life sciences and its implications for biosecurity (Sandbrink, 2023).

Moreover, as AI governance becomes more important, there is growing concern that excessive regulation could put on hold beneficial advancements and innovation, potentially making difficult international collaboration and limiting access to AI tools for positive applications (InterAcademy Partnership, 2024). The EU has a heterogeneous biosecurity landscape, shaped by varying national understandings, priorities, capabilities, and resources. This means the EU can play a supporting and guiding role to ensure consistent and effective action across all Member States. Under this context, proposed actions include:

- Adopt an **adaptive EU governance approach, where AI developments and biological risks can be monitored and polices adapted accordingly** (Carter et al., 2023a).
- Establish a **structure for reporting to competent authorities about accidents or awareness on potential misuse of technology** (Urbina, Lentzos, Invernizzi, & Ekins, 2022). Given biosecurity's global nature, **the EU could play a key role in developing a common reporting framework to facilitate information exchange among Member States** and encourage broader international adoption.

The four recommended areas detailed above establish a basis for advanced AI-driven monitoring and surveillance of biothreats as well as enforce compliance of biosecurity measures. For example, to effectively deploy AI technologies for biosecurity monitoring and surveillance, there must be parallel investments in collaborative networks and knowledge-sharing mechanisms across disciplines and borders. A targeted research agenda on AI for biosecurity can drive innovation in genetic sequence screening and real-time biological threat detection. Moreover, developing common standards and guidelines will ensure that AI-based detection systems operate reliably and consistently across Member States. Moreover, to implement advanced compliance measures, such as AI-supported supply chain tracking or automated regulatory audits, requires a regulatory framework that reflects the diverse resources and capacities across EU Member States. This highlights the need for tailored legislative approaches that account for national differences while maintaining alignment with broader EU regulations like the AI Act. Education and training programmes on AI ethics and biosecurity will also be essential to support regulatory bodies and research institutions in adopting AI tools for compliance monitoring.

Summarising, **the EU can leverage AI to enhance biosecurity by promoting cross-border cooperation, fostering research and innovation, bridging expertise gaps, and strengthening regulatory frameworks**. These actions in turn could contribute to improving monitoring capabilities and enforcing compliance, which would help the EU proactively address emerging biosecurity challenges while maintaining ethical and legal safeguards.

3.2. Final conclusions of the study

Addressing cross-border health threats is a top priority for Europe, as outlined in the European Health Union, the EU's legal framework aimed at enhancing prevention, preparedness, surveillance, risk assessment, early warning, and response. In this context, HERA is driving key initiatives to promote advanced research and development of technologies to improve preparedness and response to novel pandemics. **The integration of AI in biosecurity offers a unique opportunity to strengthen the EU's defences against both natural and man-made biological risks**.

This study has examined the literature and interviewed experts in the field to address three main objectives:

1. Assess AI's potential to enhance screening of synthetic nucleic acid orders;
2. Identify and evaluate AI-based tools and methodologies that enhance the detection and attribution of genetic engineering activities;
3. Evaluate the opportunities and risks posed by AI technologies toward biosecurity.

The main findings from the study are described below.

First, **the study team identified the seven main tools used for screening synthetic nucleic acid orders**. The underlying methodologies of most of these tools are not always publicly available. However, the available information pointed to four of them being enhanced by the use of AI techniques. The role of AI in these tools is mainly the management and refinement of reference datasets, optimisation of sequence comparison algorithms through Machine Learning, or the prediction of structure and functionality of the biological entities encoded by the sequences analysed. It is expected for AI to further contribute to the development of screening methodologies, gaining independence from traditional alignment-based methods that rely on limited repositories of well-known pathogens.

Second, **the study identified two AI-based tools for detecting genetic engineering and six for genetic engineering attribution**. The most employed AI technique in these tools is Deep learning, with some tools employing traditional clustering methods and Random Forest techniques. The literature points to an emerging field where, developing further these capabilities will be crucial for both preventing and early detecting bioweapon attacks and attributing responsibility in the event of a biosecurity breach. However, in comparison to the screening of synthetic nucleic acid orders, this field is still under development with most of the solutions developed for research purposes and not yet suited for real-world wide application.

Third, the study team evaluated the key benefits and risks of integrating AI-enabled tools into EU biosecurity measures. **AI offers key benefits for biosecurity with an impact in the areas of biosurveillance, early warning systems, outbreak monitoring and biothreat response and mitigation.** Large Language Models (LLMs) can automate biosecurity tasks, improving efficiency in tasks like data gathering or report reviewing and summarising. AI can also improve the monitoring and surveillance of outbreaks through enhanced pattern recognition in surveillance data. AI-based Biodesign Tools (BDTs) can expedite responses to biothreats by improving the design of medical countermeasures; Tools like AlphaFold2 facilitate protein structure prediction, advancing drug and vaccine development. While AI has the potential to bolster biosecurity, many applications are still evolving and face challenges, particularly concerning information quality and resource utilisation. The integration of AI also brings risks and dual-use concerns. **The two primary risks identified are the increased public access to and understanding of technical knowledge via LLM**, which could empower malicious actors, and the heightened ability to design harmful biological agents using BDTs that may evade standard screening and detection protocols. As most of the risks are closely linked to the opportunities AI offers to biosecurity, it is crucial to implement policy actions to carefully manage the current and emerging risks stemming from the advancement of rapidly evolving fields like synthetic biology and AI. These policies are essential to safeguard biosecurity while ensuring that innovation and growth in synthetic biology and AI continue to flourish.

Based on the literature findings and the targeted interviews with experts in the field, **this report drafts several recommendations on the role that the EU can take in supporting and guiding Member States in the implementation of a biosecurity framework enhanced by AI**. The main areas of recommendations are:

1. Promote multidisciplinary and cross-border collaboration in the fields of AI and biosecurity.
2. Support Research, Innovation and Capacity building in AI for Biosecurity.
3. Develope Standards and Guidelines for the use of AI and biotechnology in Life Sciences.
4. Establish rules and legislation for AI in the life sciences to improve EU's Biosecurity.

The EU can leverage AI to enhance biosecurity by fostering cross-border collaboration, promoting research and innovation, and strengthening regulatory frameworks. These initiatives can enhance the monitoring abilities, ensure compliance, and assist the EU in tackling new biosecurity threats while upholding ethical and legal standards. By adopting these actions, the EU could increase their support to Member States, enabling them to establish or reinforce a proactive and robust biosecurity framework that responds to existing risks and prepares for future challenges.

An important learning is the importance of accompanying **technical developments by political and legislative actions to support the implementation and adoption of these screening practices globally**. An example is the US Executive Order on the Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence (The White House, 2023), which mandated the screening of synthetic nucleic acid orders for those projects receiving US national funds. Although the Executive Order only covered activities funded by US national funds, it is a good example of how regulations can reinforce the adoption of existing guidelines and best practices to reinforce biosecurity and make it a subject of current interest in international discussions and among a variety of different stakeholders.

By embracing innovation and fostering collaboration, the EU can lead the way in leveraging AI to strengthen biosecurity, protect public health, and enhance global resilience against emerging biological threats. The integration of AI into the biosecurity framework presents a transformative opportunity to enhance the EU's capacity to anticipate, detect, and respond to emerging biological threats. By leveraging AI's potential for data analysis, early warning systems, and risk assessment, the EU can build a more proactive and resilient biosecurity infrastructure. However, this must be accompanied by robust ethical guidelines, regulatory frameworks, and international and multidisciplinary cooperation to ensure that AI is used responsibly and effectively. As biosecurity challenges continue to evolve, the EU's commitment to innovation, collaboration, and adaptive governance will be key to safeguarding public health and reinforcing global security. The future of biosecurity depends on the ability to balance technological advancement with careful oversight, ensuring that AI remains a powerful tool in the fight against biological risks.

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Annex: Methodology

This section outlines the methodology followed in this report for the literature review on the use of AI-based tools for enhancing the EU biosecurity framework. Moreover, this section also describes the methodology used to gather additional inputs on the topic from key stakeholders, covering from the selection of stakeholders up to the analysis of the inputs gathered.

Scoping literature review

The focus of the scoping literature review was to map and analyse literature reporting on AI's potential to enhance DNA synthesis screening in Europe to prevent access to bioweapon components and the creation of biological weapons, the role of AI in advancing genetic engineering detection and attribution, and how AI-enabled tools can contribute to the overall EU biosecurity framework.

A literature search was conducted following the PRISMA guidelines, which were designed to help reviewers transparently report why the review was done, what the authors did, and what they found (Page et al., 2021). The literature search was performed on the academic database PubMed. Other databases with a specific focus on AI research were also considered (i.e., IEEE and ACM Digital Library) but were discarded as they only contributed with publications already identified in the PubMed database. Box 6 presents the search string used for this literature review. The search strings were developed based on the key research questions of the study (see Box 6) and refined through discussions among the study team. The strings were tested to ensure they produced valid and reliable outcomes.

The study team established the inclusion criteria: studies published in the last 10 years (January 2015- August 2024) that were open source, with full text available and written in English. The exclusion criteria were also defined, with studies excluded when 1) not peer-reviewed, 2) outside the research scope, or 3) irrelevant to the research questions.

Box 6 *Research Questions and Query string used*

Research Question #1

How can AI-enabled screening techniques enhance the detection of cases of suspicious orders of synthetic DNA, possibly aimed at the creation of biological weapons?

Research Question #2

How can ML algorithms be used to improve genetic engineering detection and attribution?

Research Question #3

How can AI tools contribute to the improvement of the European and global biosecurity framework?

Query in PubMed:

("artificial intelligence"[Title/Abstract] OR "AI"[Title/Abstract] OR "ML"[Title/Abstract] OR "Machine Learning"[Title/Abstract] OR "Deep Learning"[Title/Abstract] OR "LLM"[Title/Abstract] OR "computational biology"[Title/Abstract] OR "health technolog*"[Title/Abstract] OR "big data*"[Title/Abstract]) AND ("biological weapon*"[Title/Abstract] OR "bioweapon*"[Title/Abstract] OR "genetic modif*"[Title/Abstract] OR "genetic engineer*"[Title/Abstract] OR "genetic synthes*"[Title/Abstract] OR "harmful genetic*"[Title/Abstract]) AND ("biosecurity"[Title/Abstract] OR "security"[Title/Abstract] OR "health security"[Title/Abstract] OR "detection"[Title/Abstract] OR "attribution"[Title/Abstract] OR "monitoring"[Title/Abstract] OR "unauthorised"[Title/Abstract] OR "screening"[Title/Abstract] OR "order"[Title/Abstract])

Hits: 72

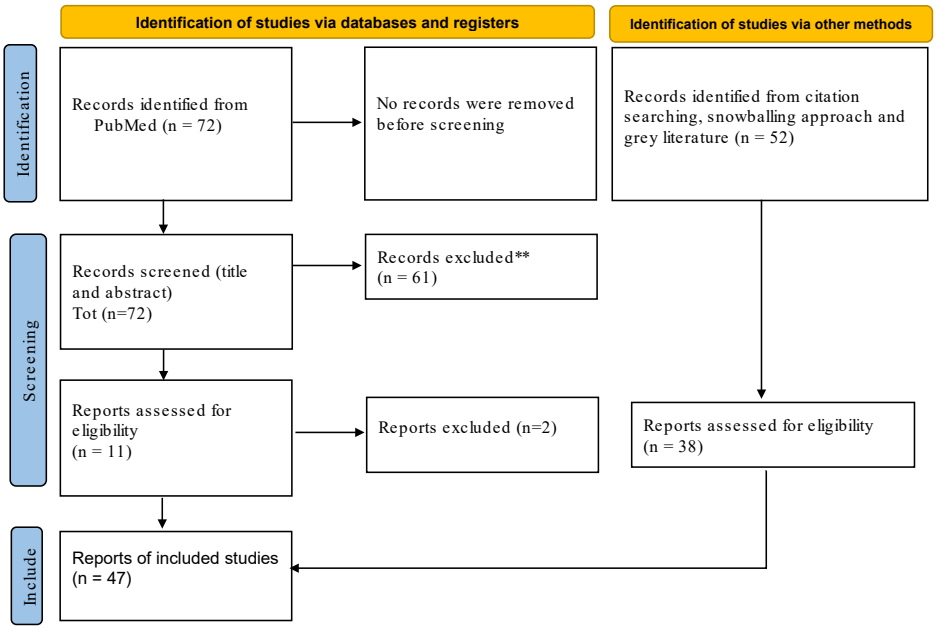
Search results (n=72) were compiled into a single file for screening. Titles and abstracts were reviewed to determine if they addressed: 1) AI-based solutions for the screening of synthetic nucleic acid orders, 2) AI-based solutions for advancing genetic engineering detection and attribution, and 3) the impact of AI-based tools on biosecurity. Publications not excluded during the title and abstract screening (n=11) underwent a rapid full-text preliminary review. Studies were included or excluded based on the exclusion criteria defined above. Nine studies were included in the review for data extraction.

Additionally, 38 key references were identified through a manual search of relevant publications and citations in included studies (snowballing approach).

Data from each 47 included studies were extracted by a single reviewer and verified for accuracy and completeness by a second independent reviewer. A narrative synthesis was performed.

The total number of records after each screening round was documented using the PRISMA flow diagram template (see Figure 1). The literature review assisted the study team in identifying authors that explored the contribution of AI-based tools to the EU biosecurity framework.

Figure 1 Prisma Flow Diagram summarising the records of the literature review.



Source: *Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71. For more information, visit: http://www.prisma-statement.org/*

Consultation activities and stakeholder mapping

The consultation activities and stakeholder mapping were carried out for six months and aimed to achieve three primary objectives:

- **Identify key experts** in AI, biosecurity and associated fields from Academia, Industry, and Government.
- **Validate and bridge research gaps**: To corroborate and fill the research gaps identified in existing literature by integrating insights from experts across academia, public health laboratories, industry, and government.

- **Validate AI methodologies and simulation methods:** To assess the validity and feasibility of AI supported methodologies identified in the literature for demand forecasting and supply chain monitoring as well as simulation methods for supply chain optimisation.

A comprehensive stakeholder mapping exercise was conducted at the outset of the study, identifying key experts across the three stakeholder categories. This mapping was validated by HERA and informed the subsequent engagement activities, which included targeted emails to request participation in specific consultation activities. The consultation activities consisted on in-depth one-on-one interviews (60 minutes long).

Prior to the consultations, three distinct interview guidelines were developed, each tailored to a specific stakeholder category (academia, industry, and government). These guidelines were reviewed and validated by HERA, ensuring comprehensive coverage of key topics while allowing flexibility for individual expertise.

Each guideline included the following sections:

- Introduction and consent for recording and minute-taking.
- Background questions on the interviewee's profile and experience in the field.
- Questions regarding the use of AI methodologies and simulation methods for demand forecasting and supply chain monitoring, considering applicability to MCM supply chain.
- Questions exploring implementation challenges (with a focus on public authorities' perspective), research bottlenecks and priorities.
- Open-ended questions for additional insights, uncovered areas, or recommendations for further consultation.

The consultation strategy envisioned a target number of participants for each category:

- Academia (N = 5)
- Industry (N = 5)
- Government (N =5)

Due to the ongoing stakeholder mapping exercise, additional stakeholders beyond the originally selected group were invited to participate in the consultation activities following a comprehensive review of their expertise. This expansion of the initially selected pool of experts enhanced the quality of input and feedback collected. Table 7 summarises the number of in-depth interviews conducted, the number of participants and their geographical regions.

Table 7 *Consultation activities of the study.*

	Number of interviews	Geographical areas represented (number of participants)
Academia	7	EU (2); UK (1); Switzerland (1); US (3)
Industry	6	US industry (5); International industry organisation (1)
Government	3	EU (2); International organisation (2)

Interviews followed a semi-structured format, allowing for adaptation based on the interviewee's specific expertise. Preliminary materials, including interview guidelines and tables of AI methodologies and simulation methods identified in the literature, were shared with interviewees in advance. Following each interview, detailed minutes were shared for review and approval before incorporating the feedback into the study.

The valuable insights gathered from the consultation activities are presented at the end of each relevant results section (Box 2-6). These findings collectively contributed to the final discussion in Section 3 of this report.

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